DiffMSR: A Multi-Semantic Graph Diffusion Model for Service Recommendation

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***Abstract*—With the rapid development of cloud computing and service computing, service recommendation systems play a crucial role in helping users efficiently filter the appropriate services. However, the sparsity of service data and the presence of noise in interactions make it extremely challenging to accurately capture user preferences. Existing service recommendation methods based on Graph Neural Networks (GNNs) primarily rely on ID aggregation, often neglecting the richness of textual semantics and are susceptible to interaction noise, resulting in suboptimal modeling of user-service relationships. Although Large Language Models (LLMs) demonstrate remarkable advantages in capturing textual semantics, current methods struggle to effectively align structural representations with textual representations, limiting improvements in recommendation performance. To address these challenges, we propose an innovative multi-semantic graph diffusion model for service recommendation, DiffMSR, which aims to align textual and structural representations while learning the generation process of interaction graphs in a denoising manner. This approach mitigates data sparsity and effectively reduces noise interference. Specifically, the model leverages LLMs to capture the textual semantic features of service descriptions and integrates them with structured semantic information from knowledge graphs. Through cross-semantic contrastive learning, it achieves heterogeneous semantic alignment. Furthermore, the model introduces a multisemantic diffusion-based generation framework, which iteratively denoises to construct high-quality user-service interaction graphs. This significantly enhances the multi-semantic awareness of user representations, thereby improving recommendation performance. Experiments on public service datasets demonstrate that DiffMSR outperforms existing state-of-the-art baseline methods, achieving improvements of 4.13% and 6.37% in recommendation accuracy and recall, respectively.**

***Index Terms*—LLM, knowledge graph, graph diffusion, service recommendation, semantic alignment.**

Received 3 April 2025; revised 8 July 2025; accepted 4 August 2025. Date of publication 8 August 2025; date of current version 9 October 2025. This work was supported in part by the CCF-Sangfor 'Yuanwang' Research Fund under Grant CCF-SANGFOR OF 2024002 and in part by the State Key Laboratory Program for Novel Software Technology, Nanjing University, P.R. China under Grant KFKT2024B52. *(Corresponding author: Buqing Cao.)*

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Digital Object Identifier 10.1109/TSC.2025.3596891

I. Introduction

With the growing popularity of Service-Oriented Architecture (SOA) and cloud computing, an increasing number of enterprises and organizations have developed diverse Web Application Programming Interfaces (API) and provided services through API-sharing platforms. This not only enhances the modularity and flexibility of software development but also enables developers to build applications with reduced time and cost [1], [2], [3]. For instance, the ProgrammableWeb.com platform offers a wide variety of publicly available API services spanning numerous application domains. The platform has shared 22,622 Web APIs, which have successfully supported developers in creating 6,467 Mashups or applications [4]. However, with the exponential growth of Web services and the increasing complexity of user demands, it has become challenging for users (i.e.developers) to quickly identify the most suitable services from a vast array of heterogeneous options. In this context, service recommendation systems have emerged as a solution. By analyzing users' historical behaviors and needs, these systems intelligently predict potential user preferences and provide personalized service recommendations. This helps developers more efficiently filter and integrate Web services, significantly improving the efficiency of application development [5].

In recent years, service recommendation techniques have undergone a significant evolution from traditional methods to deep learning approaches [6]. In particular, graph neural network (GNN)-based methods [7], [50], which construct and utilize multirelational service networks, can effectively capture functional dependencies and invocation relationships between services, becoming the main direction of current research. However, existing GNN models overly rely on service ID information for feature learning, mainly aggregating neighboring node information through message passing mechanisms while neglecting the rich textual semantic information contained in service description documents, such as API interface specifications and functional descriptions (as shown in Fig. 1). This simplification limits the model's in-depth understanding of service characteristics and the effectiveness of representation learning. Furthermore, most existing service recommendation models depend on high-quality labeled data (i.e., observed user interactions) for supervised training [8], [9], but the sparsity of interactions in real-world scenarios restricts accurate modeling of user preferences. Moreover, noise in interaction data (such as information bias [10]) is easily amplified in sparse environments,

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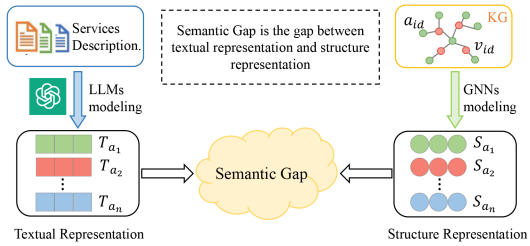

Fig. 1. An illustration of the semantic differences between GNN-based representations and textual representations.

and traditional GNN models [11] often fail to effectively distinguish noise and outliers, treating all nodes and their neighbors equally, thereby limiting the models' performance.

Regarding the above challenges, existing research mainly explores in two directions. In terms of feature enhancement, the generalizability of models is improved by introducing self-supervised learning techniques. For example, SGL [12] utilizes random perturbations to generate multiview representations, KGCL [13] adopts a random relation dropout enhancement scheme to generate different views of the knowledge graph structure, while MCCLK [14] and HCCF [15] discover implicit associations by analyzing high-order connections. However, these methods are mainly limited to enhancements at the graph structure level. In terms of semantic understanding, functional or topic features are extracted from Web Services Description Language (WSDL) documents, and services are clustered according to similarity to recommend the best matching services to user needs [16], [17]. Nevertheless, these methods fail to fully exploit the more fine-grained and rich potential semantic information in service description documents. In recent years, with the breakthrough of large language models (LLMs) such as GPT-4 [18] in understanding technical documents [19], [20], researchers have begun to explore LLM-based service feature extraction methods [21]. For instance, RLMRec [22] and Graph Translator [23] propose a multitask learning framework combined with LLMs and an instruction fine-tuning strategy, respectively. Despite some progress made by these works, effectively integrating textual semantic information with graph structural features in sparse and noisy environments still faces two key challenges:

1) Semantic gap in multi-semantic data fusion. A significant semantic gap exists between unstructured textual descriptions capturing functional and contextual semantics, and structured knowledge graphs encoding relational topologies (as illustrated in Fig. 1). This representational inconsistency hinders effective information integration and semantic alignment. Effectively incorporating service text semantics with multilevel associative semantics in knowledge graphs to mine users' potential preferences is an important direction for improving the robustness and accuracy of recommendation systems.
2) Impact of noise issues. Current self-supervised learning-based methods often rely on simple random enhancements or intuitive view alignment, which may introduce irrelevant noise information, including enhanced self-supervised signals generated by popularity bias. The performance of recommendation systems is easily affected by these noisy data, negatively impacting the precision of the recommendation.

To address the above challenges, we propose a Multi-Semantic Graph Diffusion Model for Service Recommendation (DiffMSR). This model first leverages LLMs to extract fine-grained semantic features from service description texts while obtaining structured relationship representations between services from the knowledge graph. On this basis, dynamic alignment of the text semantic space and graph structure semantic space is achieved through a designed cross-semantic contrastive learning strategy. Specifically, the model introduces a semantically aware attention mechanism to adaptively adjust the fusion weights according to the complementarity of different semantic representations. To mitigate the noise issue, we adopt a diffusion model-based generative framework to learn the generation of the user-service interaction graph through an iterative denoising process. This process first corrupts the original interaction graph by gradually adding Gaussian noise, and then designs a semantically aware denoising network that integrates multisemantic information into the denoising process to progressively restore high-quality interaction relationships. In this manner, the model can effectively identify and filter out noisy features while preserving user true preference information, thereby significantly improving the performance of service recommendation.

In summary, the contributions of this paper can be summarized as follows:

- We propose a novel multi-semantic graph diffusion model for service recommendation, DiffMSR, which leverages the semantic understanding capabilities of LLMs and the structured representations of knowledge graphs. By employing cross-semantic contrastive learning, the model achieves effective alignment and integration of heterogeneous semantic spaces.
- In the DiffMSR framework, we design a semantic-aware generative mechanism based on diffusion models. Through an iterative denoising process and a dynamic semantic fusion strategy, this mechanism effectively reduces noise interference and optimizes the representations of users and services, thereby enhancing service recommendation performance.
- Extensive experiments conducted on real-world datasets validate the effectiveness of the proposed DiffMSR framework. The results demonstrate that DiffMSR significantly outperforms various state-of-the-art baselines, while effectively addressing the challenges of data sparsity and noise in semantic-aware collaborative relationship modeling.

The remainder of this paper is structured as follows. Section II briefly reviews related work, Section III formulates the problem, while the proposed methodology is discussed in Section IV. The analysis of numerical experiments is presented in Section V, followed by conclusions and future research directions in Section VI.

II. Related Works

In this work, we present a graph diffusion service recommendation framework that combines large language models and GNN, which is closely related to GNN-based service recommendation, contrastive learning-based service recommendation, and large language model recommendation.

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A. GNN-Based Service Recommendation

In recent years, researchers have actively explored the application of GNNs [24] to tackle the challenges of service recommendation. GNN-based models capture higher-order collaborative relationships by integrating information about various services, constructing networks that represent services and their relationships, and performing message passing over graph structures [11], [49]. These models enable the detailed analysis of service connections, uncovering correlations and dependencies to provide more accurate recommendations. For instance, Xiong et al. [25] proposed a deep hybrid collaborative filtering approach for service recommendation, utilizing a multi-layer perceptron to model the complex invocation relationships between Mashups and services in Web service recommendations. Recognizing the varying importance of service relationships, Cao et al. [26] constructed a complex service relationship network with composite and annotated relationships and employed a self-attention mechanism to determine the weights of neighboring nodes in the service network. To address data sparsity and cold-start issues, some researchers have integrated knowledge graphs (KGs) as auxiliary edge information to construct service knowledge networks and employed network representation techniques for service recommendation [27]. For example, Wang et al. [28] built a knowledge graph based on Mashup-API co-invocation relationships and service category attributes. They designed an entity-biased random walk mechanism to dynamically adjust the weights of service nodes according to preferences in different neighborhoods (API or Mashup), improving recommendation accuracy. Similarly, Mezni et al. [29] developed context-aware service knowledge graphs containing multi-relational information (e.g., user, service, and context) relevant to service recommendations. They proposed a first-order and subgraph-aware embedding method based on extended recursive neural networks to achieve effective context-aware knowledge graph representations.

B. CL-Based Service Recommendation

In recent years, the advancements of self-supervised learning in the fields of natural language processing (NLP) and computer vision (CV) have garnered significant attention in recommender system research [30], [31]. Contrastive learning has demonstrated promising results by extracting valuable information from unlabeled interactions and leveraging multi-view comparisons to align node representations. For instance, Procopio et al. [32] employed LightGCN as a backbone encoder and generated enhanced interaction views using three operations: node deletion, edge dropping, and random walks, significantly improving recommendation performance. Yun et al. [33] introduced random noise to perturb node representations, simplifying the graph augmentation process. Cai et al. [34] proposed a graph augmentation strategy based on singular value decomposition, effectively capturing global collaborative signals. Choi et al. [35] constructed context-enhanced graphs incorporating geolocation information and random edge drops, and combined graph convolutional networks with contrastive learning techniques to extract embedded representations of users and services from these augmented graphs, leading to improved service recommendation performance. In addition to these approaches, some methods design cross-scale comparisons between entities of different levels. For example, Zou et al. [14] applied contrastive learning across three views at both local and global levels, enabling the mining of comprehensive graph features and structural information in a self-supervised manner. Yu et al. [36] leveraged contrastive learning to extract latent features of applications and services by augmenting sparse interactions. They designed invocation views and text views, integrating local and global contrastive learning to achieve more accurate representations of applications and services.

C. LLM-Based Recommendation

Large Language Models (LLMs) [37] have made remarkable strides in text processing and reasoning capabilities within the field of NLP. An increasing number of studies have leveraged these models to effectively capture and model user preferences [38]. A key approach involves using LLMs as enhancers, where the core idea is to utilize LLMs to refine node embeddings and subsequently integrate these enhanced embeddings into the graph structure learning process of GNNs, thereby providing a stronger foundation for downstream predictions. For instance, Wei et al. [21] employed LLMs for graph enhancement by improving both interaction relations and side information, utilizing LightGCN to encode the enhanced graphs and boost recommendation performance. Similarly, Chen et al. [39] proposed an active node selection strategy, where nodes were annotated using LLM-generated content. These annotations were then used to enhance GNN training, with LLM confidence scores employed to accurately evaluate annotation quality and optimize node selection. Meanwhile, Ren et al. [22] integrated auxiliary textual signals to analyze users and items using LLMs, capturing the complex semantic features of user behaviors and preferences.

However, while LLM-based approaches are able to deeply analyze and encode service description texts using their powerful text comprehension capabilities, they still have limitations in capturing information about the structure of the service network and have difficulty in bridging the gap between different spatial representations through direct text alignment. In addition, as mentioned in the previous section, there is still room for improvement in mitigating the effects of noise in existing methods. To this end, we propose a service recommendation framework that fuses multi-semantic graph diffusion and denoising alignment to achieve more effective recommendation results.

III. Problem Formulation

Collaborative graphs with multi-semantic features: Building on the successful application of GNNs in collaborative filtering, DiffMSR effectively leverages graph structure data to enable the fusion of multi-semantic features in service recommendation systems. In this model, the user-service interaction is represented as a graph $\mathcal{G} = (u, s) \mid u \in \mathcal{U}, s \in S$, where u and s denote the sets of users and services, respectively, and the edge (u, s) indicates that user u has interacted with service s. To enrich the user-service interaction graph with diverse semantic features, including textual and structural semantics, specific semantic feature vectors are introduced for each service $\widehat{F}_s = \hat{f}_s^1, \ldots, \hat{f}_s^m, \ldots, \hat{f}_s^{|\mathcal{M}|}$. Each vector $\widehat{f}_s^m \in \mathbb{R}^{d_m}$ represents a semantic feature of service s corresponding to a semantic type m, which belongs to the set of semantics $\mathcal{M}$. Additionally, d_m denotes the dimensionality of these semantic feature vectors. The related symbols are described in the Table I.

Task Formulation: Our goal is to develop a multi-semantic service recommendation model that accurately captures the

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TABLE I
NOTATION

Symbol	Description
$\mathcal{A}$, $\overline{\mathcal{A}}$	Interaction graph and its normalized form
$\mathcal{A}_m$, $\overline{\mathcal{A}}^m$	Semantic-aware graph and its normalized form
$\mathcal{N}_u^m$, $\mathcal{N}_s^m$	Neighbors of u, s in $\mathcal{A}_m$
E^u, E_m^s	User feature, and service semantic feature of type m
t	Diffusion step, where $t \in \{1, \ldots, T\}$
α_0, α_t	Original and noise-injected interaction at step t
Z_l^m, $\bar{Z}^m$	Layer-l and aggregated embeddings in type m
$\hat{z}^m$, k_m	Semantic representation and its weight
h_u, h_s	Final user/service embeddings
$\hat{y}_{u,s}$	Predicted preference score for user u and service s
λ_0, λ_1, λ_2	Loss weights: MSI, contrastive, regularization

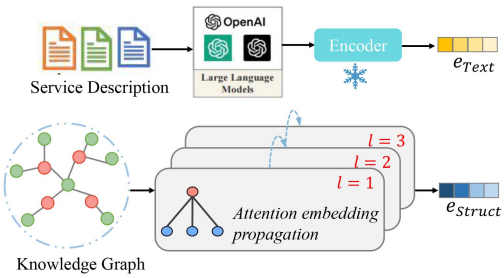

Fig. 2. Multi-semantic embedding module.

relationship between a user and a service while taking into account the multi-semantic features of the service. Specifically, our goal is to learn a function f that predicts the likelihood that a user u chooses a service s. The prediction is based on a multi-semantic interaction graph $\mathcal{G}^M = (\mathcal{G}, \{F_s, s \in S\})$as input, which is denoted as $\hat{y}_{u,s} = f(\mathcal{G}^M)$.

IV. METHODOLOGY

In this section, we present the technical design of our proposed DiffMSR model and its overall architecture, as illustrated in Fig. 3. The model is composed of four key modules: a multi-semantic embedding module, a graph diffusion module, a multi-graph aggregation module, and a cross-view comparison enhancement module. These components work together to capture various semantic relationships in user-service interactions, ensuring effective alignment between textual and structural representations. This alignment facilitates the seamless fusion of high-quality semantic information, thereby enhancing the accuracy and performance of service recommendations.

A. Multi-Semantic Embedding

We leverage the knowledge base and reasoning capabilities of the LLM to generate spatially uniform and information rich service attributes to enhance semantic embedding for improving the performance of service recommendation models. In the LLM-based side information enhancement framework, service information is semantically extended and encoded as features as model inputs. The accuracy of the service representation is crucial for a deeper understanding of the semantic preferences behind the user's interaction with the service. Since service descriptions often involve complex technical expressions such as API specifications, parameter semantics, and functional dependencies, LLMs demonstrate superior capabilities in capturing such fine-grained and domain-specific semantics, making them particularly suitable for service semantic modeling. To achieve this goal, we adopt the text-embedding model provided by OpenAI to extract high-quality semantic representations from service description documents to provide reliable semantic support for service recommendation. Specifically, the process of service text-side information enhancement based on LLM is as follows:

$$\mathbb{A}_s = \mathrm{LLM}\left(\mathbb{P}_s^S\right) \rightarrow f_{\mathcal{G},s} = \mathrm{LLM}\left(\mathbb{A}_s\right). \tag{1}$$

where $f_{\mathcal{G},s} \in \mathbb{R}^{d_{LLM}}$ is an LLM-enhanced service feature with LLM's hidden dimension d_{LLM}. The textual cue $\mathbb{P}_S^S$ is used for attribute refinement of the service. $\mathbb{A}_S$ denotes the textual attributes of the service that have been generated and encoded as feature $f_{\mathcal{G},s}$ using the embedding capability of $\mathrm{LLM}(\cdot)$. We utilize Knowledge Graph (KG) to extract deep structural semantic associations between services, as shown in Fig. 2. This enhances the model's ability to understand structural semantic features by mining complex relationships between services. Knowledge Graph not only provides rich context for service descriptions, but also lays the foundation for capturing potential associative relationships between services. Based on this, we introduce the attention mechanism of Knowledge Graph Attention Network for flexibly adjusting the weights of inter-service associations. The mechanism dynamically assigns weights according to the importance of semantic relationships, highlighting key semantic features while reducing the interference of secondary information, thus improving the accuracy and robustness of service recommendation. Formally, the process of obtaining the KG-based service structure information representation is as follows:

$$\pi(h, r, b) = \mathrm{softmax}\left\{(W_r e_b)^\top \tanh\left(W_r e_h + e_r\right)\right\}. \tag{2}$$

$$e_{\mathcal{N}_h} = \sum_{(h,r,b)\in\mathcal{N}_h} \pi(h, r, b) e_b. \tag{3}$$

Here, $\pi(h, r, b)$ represents the attenuation factor that controls each propagation over the edges (h, r, b), specifically indicating the effective information propagated from the tail node b to the head node h, conditioned on the relation r. The factor $\pi(h, r, b)$ is implemented through a relational attention mechanism, with the $\tanh(\cdot)$ function chosen as the nonlinear activation function. This ensures that the attention score depends on the distance between e_h and e_b in the relation r space. To simplify the computation, we apply an inner product operation on these representations and normalize the resulting attention scores using the function. Furthermore, to better characterize the structural properties of the triple set $\mathcal{N}_h = \{(h, r, b) \mid (h, r, b) \in \mathcal{G}^{kg}\}$ and capture the first-order connectivity information of the entity h, we compute the linear combination $e_{\mathcal{N}_h}$ of the self-network. This approach allows for a more comprehensive modeling of the structural semantics of the service nodes.

B. Graph Diffusion Model

Inspired by diffusion models [40], [41], which effectively preserve essential data patterns during the generation process, we propose an innovative approach DiffMSR for multi-semantic service recommendation. This framework generates user-service interaction graphs enriched with semantic information through a multi-semantic graph diffusion module, which enhances the modeling of user preferences while mitigating interference from irrelevant or noisy features in service recommendation systems. Specifically, we leverage a large language model to refine the semantic information of service texts and

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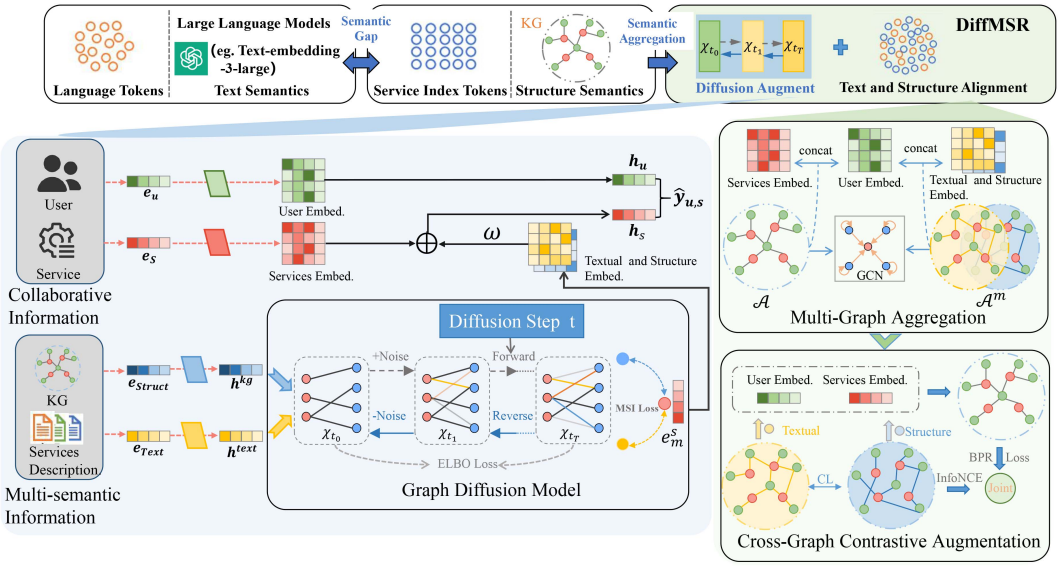

Fig. 3. The framework of DiffMSR.

integrate it with the structural semantics of the knowledge graph. This integration uncovers deep associations between services, enriching the multi-semantic features of the interaction graph. Furthermore, we introduce a probabilistic denoising diffusion model to enable enhanced semantic awareness. By progressively corrupting and restoring user-service interaction relationships, this model facilitates semantic fusion during iterative learning, ensuring the effective integration of multi-semantic information while mitigating the impact of noise interference.

To better capture multi-semantic information, we introduce a semantic-aware signal injection mechanism. This mechanism dynamically incorporates diverse semantic feature vectors into the interaction graph generation process. It guides the recovery of interactions and ensures the effective integration of semantic information.

Unlike other semantic feature models that employ random Gaussian noise for inverse generation, we utilize diffusion training for user-service interaction prediction. Specifically, we first disrupt α_0 through the forward diffusion process $\alpha_0 \to \alpha_1 \to \cdots \to \alpha_{T'}$, and then set $\widehat{\alpha}_T = \alpha_{T'}$. Subsequently, reverse denoising is performed $\widehat{\alpha}_T \to \widehat{\alpha}_{T-1} \to \cdots \to \widehat{\alpha}_0$ to recover the original interactions. The reverse denoising process is defined as $\widehat{\alpha}_{t-1} = \mu_\theta(\widehat{\alpha}_t, t)$, which ensures accurate predictions. Finally, the recovered $\widehat{\alpha}_0$ is used to reconstruct the user-service graph. Specifically, for a given user u, we define $\hat{a}_u = \widehat{\alpha}_0 = [\hat{a}_u^0, \hat{a}_u^1, \ldots, \hat{a}_u^{|S|-1}]$. We select the top-k interactions $\hat{a}_u^i (i \in [0, |S|-1], i \in \mathcal{E}' and |\mathcal{E}| = k)$ and add u and the service s $\in S$ among the top-k interactions as edges. The resulting user-service graph for semantic m is denoted as $\mathcal{A}_m$.

Forward Graph Diffusion Process: Consider the inter-actions between user u and the service set S, represented as $a_u = [a_u^1, a_u^2, \ldots, a_u^{|S|-1}]$, where $a_u^s = 1$ or 0 indicates whether user u has interacted with service s. The forward diffusion process starts with the initialization $\alpha_0 = a_u$, and over T steps, adds Gaussian noise step by step to simulate the Markov chain $\alpha_{1:T}$. The transition parameters at step t are defined as follows:

$$q\left(\alpha_t \mid \alpha_{t-1}\right) = N\left(\alpha_t; \sqrt{1-\beta_t}\alpha_{t-1}, \beta_t S\right). \tag{4}$$

where $\beta_t \in (0,1)$ controls the noise scale at each step $t \in \{1, \ldots, \mathrm{T}\}$, and N denotes the Gaussian distribution. As $T \to \infty$, α_T approximates a standard Gaussian distribution. Using reparameterization techniques and the independence property of Gaussian noise, we can directly derive α_t from α_0. Formally, the process can be expressed as:

$$q\left(\alpha_t \mid \alpha_0\right) = N\left(\alpha_t; \sqrt{\bar{\gamma}_t}\alpha_0, \left(1-\bar{\gamma}_t\right) S\right). \tag{5}$$

To control the noise level in $\alpha_{1:T}$, we introduce two parameters, $\gamma_t = 1 - \beta_t$ and $\bar{\gamma}_t = \prod_{t'=1}^{t} \gamma_{t'}$, and rewrite α_t as: $\alpha_t = \sqrt{\bar{\gamma}_t}\alpha_0 + (1-\bar{\gamma}_t)\epsilon$, where $\epsilon \sim N(0, S)$. We also introduce a linear noise adjustment mechanism to dynamically control the noise level in $\alpha_{1:T}$:

$$1-\bar{\gamma}_t = \partial \cdot \left[\gamma_{\min} + \frac{t-1}{T-1}\left(\gamma_{\max} - \gamma_{\min}\right)\right], \quad t \in \{1, \ldots, T\}. \tag{6}$$

where $\partial \in [0,1]$ adjusts the overall noise intensity, $\gamma_{\max}$ and $\gamma_{\min}$ (both in the range (0,1)) are the upper and lower bounds of the noise levels, respectively.

Reverse Graph Diffusion Process: The objective of DiffMSR is to eliminate the noise introduced in α_t and recover α_{t-1} during the reverse steps. This process enables multi-semantic diffusion to effectively capture subtle variations in complex generative processes. Starting from α_T, the denoising transition steps progressively restore user-service interactions. The reverse procedure unfolds as follows:

$$p_\theta\left(\alpha_{t-1} \mid \alpha_0\right) = N\left(\alpha_{t-1}; \mu_\theta\left(\alpha_t, t\right), \Sigma_\theta\left(\alpha_t, t\right)\right). \tag{7}$$

Here, $\mu_\theta(\alpha_t, t)$ and $\Sigma_\theta(\alpha_t, t)$ represent the mean and covariance values of the predicted Gaussian distributions, respectively. These values are generated by two neural networks with learnable parameters θ.

Semantic-aware optimization of graph diffusion: The central goal in training diffusion models is to guide the reverse graph diffusion process. To achieve this, we need to optimize the Evidence Lower Bound (ELBO) of the negative log-likelihood of the observed user-service interactions α_0. The specific formulation is as follows:

$$\begin{aligned}\mathcal{L}_{\text{elbo}} &= \mathbb{E}_{q(\alpha_0)}\left[-\log p_\theta\left(\alpha_0\right)\right] \\ &\leq \sum_{t=0}^{T} \mathbb{E}_q\left[\mathcal{L}_t\right], \quad t \in \{0, \ldots, T\}.\end{aligned} \tag{8}$$

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There are three different cases for $\mathcal{L}_t$, as shown below:

$$\mathcal{L}_t = \begin{cases} -\log p_\theta(\alpha_0 \mid \alpha_1), & \text{if } t = 0, \\ D_{\mathrm{KL}}(q(\alpha_T \mid \alpha_0) \parallel p(\alpha_T)), & \text{if } t = T, \\ D_{\mathrm{KL}}(q(\alpha_{t-1} \mid \alpha_t, \alpha_0) \| \\ \qquad p_\theta(\alpha_{t-1} \mid \alpha_t)) & \text{if } t \in \{1, 2, \ldots, T-1\}. \end{cases} \tag{9}$$

Where $\mathcal{L}_0$ represents the negative reconstruction error computed over α_0, $\mathcal{L}_t$ is a fixed constant that can be ignored during optimization since it does not contain trainable parameters, $\mathcal{L}_t(t \in \{1, 2, \ldots, T-1\})$ is used to regularize $p_\theta(\alpha_{t-1} \mid \alpha_t)$ to keep it consistent with the tractable real-world transition $q(\alpha_{t-1} \mid \alpha_t, \alpha_0)$.

To optimize the graph diffusion process, we design a neural network that performs the reverse denoising operation. Specifically, the goal of $\mathcal{L}_t$ is to minimize the KL divergence, allowing $p_\theta(\alpha_{t-1} \mid \alpha_t)$ to approximate a tractable distribution $q(\alpha_{t-1} \mid \alpha_t, \alpha_0)$, as shown in (9). By applying Bayes' theorem, $q(\alpha_{t-1} \mid \alpha_t, \alpha_0)$can be rewritten in the following form:

$$q\left(\alpha_{t-1} \mid \alpha_t, \alpha_0\right) \propto N\left(\alpha_{t-1}; \tilde{\mu}(\alpha_t, \alpha_0, t), \sigma^2(t) S\right). \tag{10}$$

$$\begin{cases} \tilde{\mu}(\alpha_t, \alpha_0, t) = \frac{\sqrt{\gamma_t(1-\bar{\gamma}_{t-1})}}{1-\bar{\gamma}_t}\alpha_t + \frac{\sqrt{\bar{\gamma}_{t-1}(1-\gamma_t)}}{1-\bar{\gamma}_t}\alpha_0, \\ \sigma^2(t) = \frac{(1-\gamma_t)(1-\bar{\gamma}_{t-1})}{1-\bar{\gamma}_t}. \end{cases} \tag{11}$$

Where $\mu(\alpha_t, \alpha_0, t)$ and $\sigma^2(t)S$ represent the mean and covariance of $q(\alpha_{t-1} \mid \alpha_t, \alpha_0)$, respectively. To simplify the calculation and ensure the stability of the training process, we ignore the learning of $p_\theta(\alpha_{t-1} \mid \alpha_t)$ in $\Sigma_\theta(\alpha_t, t)$ and directly set $\Sigma_\theta(\alpha_t, t) = \sigma^2(t)S$. Equation (12) is a simplified formulation derived from (9) under the Gaussian assumption. Consequently, the loss function can be expressed as:

$$\mathcal{L}_t = \frac{1}{2\sigma^2(t)} \left\|\mu_\theta(\alpha_t, t) - \tilde{\mu}(\alpha_t, \alpha_0, t)\right\|_2^2. \tag{12}$$

The objective of this loss function is to make $\mu_\theta(\alpha_t, t)$ approximate $\tilde{\mu}(\alpha_t, \alpha_0, t)$. According to (11), we can similarly decompose $\mu_\theta(\alpha_t, t)$ to further optimize the model:

$$\mu_\theta(\alpha_t, t) = \frac{\sqrt{\gamma_t(1-\bar{\gamma}_{t-1})}}{1-\bar{\gamma}_t}\alpha_t + \frac{\sqrt{\bar{\gamma}_{t-1}(1-\gamma_t)}}{1-\bar{\gamma}_t}\hat{\alpha}_\theta(\alpha_t, t). \tag{13}$$

$$\mathcal{L}_t = \frac{1}{2}\left(\frac{\gamma_{t-1}}{1-\bar{\gamma}_t} - \frac{\gamma_t}{1-\bar{\gamma}_t}\right) \|\hat{\alpha}_\theta(\alpha_t, t) - \alpha_0\|_2^2. \tag{14}$$

In the optimization process, we use (14) to calculate the loss function $\mathcal{L}_t$. This equation is based on the predicted value $\widehat{\alpha}_\theta(\alpha_t, t)$, which is derived from α_t and time step t. We employ a neural network, specifically a multi-layer perceptron (MLP), that takes the embeddings α of t and as inputs to realize the prediction $\widehat{\alpha}_\theta(\alpha_t, t)$.

$$\mathcal{L}_0 = \|\hat{\alpha}_\theta(\alpha_1, 1) - \alpha_0\|_2^2. \tag{15}$$

For the initial time step, $\mathcal{L}_0$ is computed using (15), where the Gaussian log-likelihood $\log p(\alpha_0 \mid \alpha_1)$ is estimated using the nonweighted term $-\|\widehat{\alpha}_\theta(\alpha_1, 1) - \alpha_0\|_2^2$. To reduce computational costs, during implementation, the time step t is uniformly sampled from $\{1, 2, \ldots, T-1\}$, resulting in the loss function $\mathcal{L}_{\text{elbo}}$, as shown in (16). It is a Monte Carlo approximation of the ELBO objective in (8), obtained by uniformly sampling a diffusion step t.

$$\mathcal{L}_{\text{elbo}} = \mathbb{E}_{t \sim \mathcal{U}(1,T)} \mathbb{E}_{q(\alpha_0)} \left[\|\hat{\alpha}_\theta(\alpha_t, t) - \alpha_0\|_2^2\right]. \tag{16}$$

To enhance the effect of DiffMSR on multi-semantic user-service interaction graph diffusion, we design a Semantic-aware Signal Injection (SSI) mechanism. This mechanism guides the diffusion process towards optimizing the service embeddings to align with multiple user-service semantics.

Specifically, we aggregate the aligned service semantic features e_m^s (detailed in Section IV) with the predicted multi-semantic user-service interaction embedding $\widehat{\alpha}_0$. Furthermore, we aggregated the observed user-service interaction α_0 into the embedding id e^S. Finally, we calculate the Mean Squared Error (MSE) loss between these two aggregated embeddings. This MSE loss, denoted as $\mathcal{L}_{msi}^m$, is used to enrich the diffusion model with semantic information and is optimized together with $\mathcal{L}_{\text{elbo}}$. Formally, the semantic MSE loss $\mathcal{L}_{msi}^m$ is shown as:

$$\mathcal{L}_{\text{msi}}^m = \|\hat{\alpha}_\theta \cdot e_m^s - \alpha_0 \cdot e^s\|_2^2. \tag{17}$$

Through this method, we inject the multi-semantic information into the diffusion model, guiding the service embeddings to conform to the desired semantic properties.

Multi-Semantic Graph Diffusion Model Reasoning: Unlike traditional models that generate semantic features through random Gaussian noise, we employ a diffusion training method to predict user-service interactions. Specifically, we first perturb the original interaction data α_0 through the corruption process $\alpha_0 \to \alpha_1 \to \cdots \to \alpha_{T'}$, then set $\widehat{\alpha}_T = \alpha_{T'}$ and execute the reverse denoising process, i.e., $\widehat{\alpha}_T \to \widehat{\alpha}_{T-1} \to \cdots \to \widehat{\alpha}_0$. During the reverse denoising process, we ignore the variance and utilize $\widehat{\alpha}_{t-1} = \mu_\theta(\widehat{\alpha}_t, t)$, for deterministic inference.

Finally, we use the obtained to reconstruct the structure of the user-service interaction graph. For user u, we compute $\hat{a}_u = \hat{a}_0 = [\hat{a}_u^0, \hat{a}_w^1, \ldots, \hat{a}_u^{|S|-1}]$. Then, we select the top-k largest values $\hat{a}_u^i (i \in [0, |S|-1], i \in \mathcal{E}$ and $|\mathcal{E}| = k)$ and add k interactions between user u and services $s \in S$. Ultimately, the resulting semantic representation of the user-service graph, denoted as $\mathcal{A}_m$, is used for subsequent recommendation tasks. This denoising process enables the model to progressively suppress noisy or less informative interactions, while preserving stable user-service preferences. The final reconstruction based on top-k denoised interactions ensures that only the most relevant associations are retained for downstream recommendation.

C. Cross-Graph Semantic Contrast Enhancement

In multi-semantic recommendation systems, different types of semantic information, such as textual semantics and structural semantics, often exhibit a certain degree of correlation. For instance, in service recommendation scenarios, textual semantics and higher-order structural semantics may jointly influence users' choices, leading to intertwined user preferences across these semantics. To effectively capture the relationships between these semantics and leverage their consistency to enhance the performance of recommendation systems, we propose two semantic-aware contrastive learning paradigms based on different anchors.

The first paradigm involves contrastive learning by treating different semantic views as anchors, while the second paradigm selects the primary semantic view as the anchor. Through this approach, we can better model the interactions between semantics, thereby improving the accuracy and efficiency of the system in capturing user preferences. Both methods aim to optimize the fusion of semantic information and enhance the overall performance of the recommendation model.

Semantics aware contrast view: We describe how cross-semantic contrast learning can be utilized to generate semantically specific user and service embedded representations. We

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adopt a GNN-based representation learning approach, which is achieved by performing information aggregation on the user-service graph $\mathcal{A}_m$ constructed by the semantic-aware generation module. First, given the original feature vector $\hat{f}^m \in \mathbb{R}^{d_m}$, we generate dimensionally-aligned service semantic features $e_m^s \in \mathbb{R}^d$ by using the following formula.

$$e_m^s = \text{Norm}\left(\text{Trans}\left(\hat{f}^m\right)\right), \quad m \in \mathcal{M}. \tag{18}$$

Where $\text{Norm}(\cdot)$ represents the normalization function, and Trans $(\cdot)$ represents a multi-layer perceptron (MLP) based mapping from $\mathbb{R}^{d_m}$ to $\mathbb{R}^d$. Subsequently, we aggregate information using the user embedding matrix $E^u \in \mathbb{R}^{U\times d}$ and the service semantic feature matrix $E_m^s \in \mathbb{R}^{S\times d}$ to obtain the aggregated semantic-aware embedding $z^m \in \mathbb{R}^d$, as follows:

$$\begin{aligned} z_u^m &= \bar{A}_{u,*}^m E^u, \\ z_s^m &= \bar{A}_{*,s}^m E_s^m, \\ \bar{A}_{u,s}^m &= A_{u,s}^m / \sqrt{|\mathcal{N}_u^m||\mathcal{N}_s^m|}. \end{aligned} \tag{19}$$

where $\overline{\mathcal{A}}^m \in \mathbb{R}^{U\times S}$ represents the normalized adjacency matrix of the semantic-aware graph $\mathcal{A}_m$. $\mathcal{N}_u^m$ and $\mathcal{N}_s^m$ denote the neighbor sets of user u and service s in $\mathcal{A}$, respectively. To capture higher-order collaborative effects under multi-semantic information, we further process the aggregated embeddings $\mathcal{A}$ through the following iterative formula:

$$Z_{l+1}^m = \overline{\mathcal{A}} \cdot Z_l^m, \quad Z_0^m = Z^m. \tag{20}$$

where Z_l^m and Z_{l+1}^m represent the embeddings of layers l and $l+1$, respectively. $\overline{\mathcal{A}}$ is the normalized adjacency matrix of graph $\mathcal{A}$, similar to the computation of $\overline{\mathcal{A}}^m$.

In our multi-layer GNN, the layer-specific embeddings are aggregated through summation pooling to generate the final output:

$$\bar{Z}^m = \sum_{l=0}^{L} Z_l^m. \tag{21}$$

where L is the number of layers in the GNN. By doing so, we effectively incorporate cross-semantic information to generate high-quality semantic-specific user and service embeddings for subsequent recommendation tasks.

Semantic-aware Contrastive Augmentation: To further optimize the semantic-aware contrastive loss, we employ two different contrastive learning methods to handle the relationships between different semantic views. One approach is to use different semantic views as anchors, while the other method selects the main view as the anchor.

Semantic view as the anchor: Based on the correlation of user behavior patterns under different semantic modes, we treat embeddings from different semantic spaces as views for contrastive learning. Specifically, we use $\bar{Z}^{m_1}, \bar{Z}^{m_2}$ for contrast, where $m_1, m_2 \in \mathcal{M}, m_1 \neq m_2$. To maximize mutual information between these views, we employ the InfoNCE loss function. Additionally, to introduce negative contrast, we use embedding pairs from different users as negative samples $(u, v) \mid u, v \in U, u \neq v$. Formally, the first contrastive learning loss can be expressed as:

$$\mathcal{L}_{\text{cl}}^{\text{user, sv}} = \sum_{m_1\in\mathcal{M}} \sum_{m_2\in\mathcal{M}} \sum_{u\in\mathcal{U}} - \log \frac{\exp\left(\text{sim}\left(\bar{z}_u^{m_1}, \bar{z}_u^{m_2}\right)/\tau\right)}{\sum_{v\in\mathcal{U}} \exp\left(\text{sim}\left(\bar{z}_u^{m_1}, \bar{z}_v^{m_2}\right)/\tau\right)}. \tag{22}$$

Here $\text{sim}(\cdot)$ represents the similarity function and τ is the temperature coefficient. Through this contrastive loss, we can maximize the consistency of similarity for positive sample pairs and minimize the similarity for negative sample pairs, effectively promoting information fusion and consistency alignment between different semantic views.

Main view as the anchor: The second contrastive learning method utilizes user behavior in different semantic views to guide and optimize the learning of the target recommendation task. Specifically, we use the embedding $\bar{H}$ from the main task as an anchor, combining information loss from different semantic views with the aim of maximizing mutual information between these views. This approach, by focusing on the main view, facilitates the effective integration of cross-semantic information, thereby enhancing the recommendation performance. Formally, the second contrastive loss function can be expressed as:

$$\mathcal{L}_{\text{cl}}^{\text{user, mv}} = \sum_{m\in\mathcal{M}} \sum_{u\in\mathcal{U}} - \log \frac{\exp\left(\text{sim}\left(\bar{h}_u, \bar{z}_u^m\right)/\tau\right)}{\sum_{v\in\mathcal{U}} \exp\left(\text{sim}\left(\bar{h}_u, \bar{z}_v^m\right)/\tau\right)}. \tag{23}$$

The contrastive learning loss for the service side, denoted as $\mathcal{L}_{cl}^{\text{service, mv/sv}}$, is calculated using a similar approach. By combining these two loss terms, we obtain the overall objective function for cross-modal contrastive learning, which is represented by $\mathcal{L}_{cl} = \mathcal{L}_{cl}^{\text{user, mv/sv}} + \mathcal{L}_{cl}^{\text{service, mv/sv}}$.

D. Multi-Semantic Graph Aggregation

To generate the final user (service) representations $\bar{h}_u, \bar{h}_s \in \mathbb{R}^d$ for prediction, we first aggregate all the semantic-aware embeddings $\hat{f}^m$ and the corresponding semantic-aware user-service graphs $\mathcal{A}^m$. Then, we perform message passing on the original user-service interaction graph $\mathcal{A}$ to explore higher-order collaborative signals.

Specifically, we first extract the aligned service semantic features from $\hat{f}^m$ using (18). Next, we apply graph aggregation $\overline{\mathcal{A}}$ and $\overline{\mathcal{A}}^m$ to obtain the semantic representation $\hat{\boldsymbol{z}}^m$ for each semantic:

$$\begin{aligned} \hat{z}_u^m &= \overline{\mathcal{A}}_{u,*} \cdot E^u + \overline{\mathcal{A}}_{u,*} \cdot \left(\overline{\mathcal{A}}_{u,*} \cdot E^u\right) + \overline{\mathcal{A}}_{u,*}^m \cdot E^u, \\ \hat{z}_s^m &= \overline{\mathcal{A}}_{s,*} \cdot E_m^s + \overline{\mathcal{A}}_{s,*} \cdot \left(\overline{\mathcal{A}}_{s,*} \cdot E^u\right) + \overline{\mathcal{A}}_{*,s}^m \cdot E^s. \end{aligned} \tag{24}$$

For all single-semantic representations $\hat{z}^m (m \in \mathcal{M})$, we aggregate the representations of each semantic by summation. Since different semantics may have varying degrees of influence on the final multi-semantic representation, we introduce learnable parameters k_m to control the weight of semantic m in the aggregated multi-semantic representations $h_u($ or $h_s)$:

$$h_u = \sum_{m\in\mathcal{M}} k_m \hat{z}_u^m, \quad h_s = \sum_{m\in\mathcal{M}} k_m \hat{z}_s^m. \tag{25}$$

Furthermore, we perform message passing on $\overline{\mathcal{A}}$ using a graph neural network to explore higher-order collaborative signals, as follows:

$$H_{l+1} = \mathcal{A} \cdot H_l, \; H_0 = H_u \text{ or } H_s. \tag{26}$$

where H_l and H_{l+1} represent the embeddings of layers l and $l+1$, respectively. Here, the size of H is adjusted through $\mathbb{R}^{S\times d}$ or $\mathbb{R}^{U\times d}$. Finally, we aggregate the embeddings from each layer using a summation pooling operation to generate the final embedding $\bar{H}$:

$$\bar{H} = \sum_{l=0}^{L} H_l + \omega \text{Norm}(H_0). \tag{27}$$

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where ω is a hyperparameter that controls the normalized weight of H_0, which helps alleviate the over-smoothing problem. Through the final embeddings, DiffMSR predicts the unobserved interactions between user u and service s using $\hat{y}_{u,s} = \bar{h}_u^T \cdot \bar{h}_s$.

E. Optimization Objectives for DiffMSR

The DiffMSR training process consists of two main parts: the training of the recommendation task and the training of the diffusion module of the multi-semantic graph. The training of the diffusion module involves the simultaneous optimization of two loss terms, ELBO loss and MSI loss. Therefore, the optimization objective of the diffusion module for semantic m can be expressed as follows:

$$\mathcal{L}_{dm}^m = \mathcal{L}_{elbo} + \lambda_0 \mathcal{L}_{msi}^m. \tag{28}$$

where λ_0 is a hyperparameter that controls the strength of the MSI loss. For the recommendation task, we introduce the bayesian Personalized Ranking (BPR) loss and combine it with the previously mentioned contrastive learning loss $\mathcal{L}_{cl}$. The adopted BPR loss $\mathcal{L}_{bpr}$ is defined as follows:

$$\mathcal{L}_{bpr} = \sum_{(u,s,j)\in\mathcal{O}} -\log \sigma\left(\hat{y}_{u,s} - \hat{y}_{u,j}\right). \tag{29}$$

where $\mathcal{O} = \{(u,s,j) \mid (u,s) \in \mathcal{O}^+, (u,j) \in \mathcal{O}^-\}$ represents the training datasets, and $\mathcal{O}^- = U \times S / \mathcal{O}^+$ denotes the unobserved user-service interactions. Based on the above definitions, the joint training objective for the recommendation task can be formulated as:

$$\mathcal{L}_{rec} = \mathcal{L}_{bpr} + \lambda_1 \mathcal{L}_{cl} + \lambda_2 \|\Theta\|_2^2. \tag{30}$$

where Θ denotes all the learnable parameters in the model, λ_1 and λ_2 are the hyperparameters controlling the learning loss of contrast and the strength of L_2 regularization, respectively.

F. Time Complexity Analysis

We analyze the time complexity of DiffMSR from three main aspects. The multisemantic embedding module incurs a complexity of $\mathcal{O}(B \times (T_{\text{text}} + K) \times d)$, where B is the batch size, T_{text} the maximum token length, K the average number of neighbors, and d the embedding dimension. For the cross-semantic contrastive learning module, the complexity is $\mathcal{O}(L \times B \times (\mathcal{I} + \mathcal{J}) \times d)$, where L denotes the number of GNN layers, and $\mathcal{I}, \mathcal{J}$ are the number of services and users, respectively. The graph diffusion process requires $\mathcal{O}(B \times \mathcal{I} \times d_{\text{diff}})$ for training and $\mathcal{O}(t \times B \times \mathcal{I} \times d_{\text{diff}})$ for inference, where t is the number of denoising steps. It is worth noting that, in practical implementations, the time complexity of the diffusion module appears to be acceptable, as it is roughly on par with that of the contrastive learning module. This implies that the overall computational cost of DiffMSR is likely to remain within a reasonable range, making it comparable to existing self-supervised recommendation methods in terms of efficiency.

V. Experiment

In this section, we present experimental results to validate the effectiveness of our proposed model, DiffMSR. To achieve this, we address the following research questions:

- RQ1: How does our proposed model perform compared to various state-of-the-art recommender systems?

Algorithm 1: The Model Training of DiffMSR.

Input: Service description set $\mathcal{D}$, knowledge graph $\mathcal{G}^{kg}$, interaction graph $\mathcal{G}$
Output: Prediction function $F(u, s \mid \Theta)$, where $\hat{y}_{u,s} = \bar{h}_u^T \cdot \bar{h}_s$.

```
Initialize model parameters Θ;
for iteration iter = 1, 2, ... do
    Sample minibatch of user-service interactions M from G;
    for each user-service pair (u, s) ∈ M do
        Forward diffusion:
            q(α_t | α_0) ← N(α_t; √γ̄_t α_0, (1 − γ̄_t) S);
        Reverse diffusion:
            p_θ(α_{t−1}|α_t) ← N(α_{t−1}; μ_θ(α_t, t), Σ_θ(α_t, t));
        Calculate ELBO loss L_elbo ;
        Extract textual and structural features: Textual features: f_{G,s} ← LLM(A_s);
        Structural features:
            e_{N_h} ← Σ_{(h,r,b)∈N_h} π(h, r, b) e_t;
        Calculate Contrastive loss L_cl ;
    Aggregate h_u and h_s:
        h_u ← Σ_{m∈M} k_m ẑ_u^m,   h_s ← Σ_{m∈M} k_m ẑ_s^m;
    Calculate Diffusion module loss L_dm^m;
    Calculate Recommendation loss L_rec ;
    Calculate Total loss L_total ← L_dm^m + L_rec;
    Update model parameters using gradient descent:
        Θ ← Θ − η Δ_θ L_total ;
Return trained model with parameters Θ;
```

TABLE II
STATISTICS OF THE EXPERIMENTAL DATASETS

Mashup-Service Interaction				Semantic Embed Dim	
Mashup	API	Interactions	Sparsity	Text	Structure
5780	1195	10626	99.85%	3072	64

- RQ2: What are the contributions of our key components to its overall performance on different datasets?
- RQ3: How do different hyper-parameters influence the results?
- RQ4: How does the user-service relation learning in DiffMSR enhance the interpretability of the generated recommendations?

A. Experiment Settings

Dataset Description: We collected network service metadata from ProgrammableWeb.com[1] , including Mashups, APIs, categories, tags, followers and description documents. After preprocessing, we extracted a subset of user-service interactions. For the knowledge graph, we retained only entities with at least two interactions. The details of the dataset statistics are shown in the Table II.

[1] https://ssy.hnust.edu.cn/download/ProgrammableWeb2022.zip

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Evaluation: To evaluate the accuracy of our DiffMSR model in top-K service recommendation tasks, we employ four widely-used metrics: Recall@K, Precision@K, Hit Ratio@K, and Normalized Discounted Cumulative Gain (NDCG@K), where K is set to 10, 15, 20, 25, 30. These evaluation metrics are extensively used to assess the performance of top-K recommendations, considering both the accuracy of the recommendations and the quality of the ranking in the recommendation list. Following the methodology adopted in previous research [11], we employ the all-ranking evaluation strategy to assess accuracy.

Baselines: To demonstrate the effectiveness of DiffMSR, we compare it with four recommendation system methods: graph convolution-based methods (LightGCN), knowledge-aware methods (CFKG, KGCN, KGAT, KGIN), self-supervised methods (SGL, KGCL, MCCLK), methods combining knowledge-awareness and generative self-supervision (KGRec, DiffKG), and multimodal methods (SMIN), as detailed below:

- *LightGCN [11]* : This method uses lightweight graph convolution operations and a stacking strategy to accumulate user and item embeddings, generating the final node representations.
- *CFKG [42]* : It combines knowledge-base embeddings for reasoning to process both structured and unstructured data. By learning from heterogeneous entity embeddings, this method enhances recommendation performance.
- *KGCN [43]* : Through multi-layer propagation and aggregation of entity neighborhood information, KGCN leverages GNNs to capture higher-order relational features in the knowledge graph, thereby enhancing the learning of entity associations.
- *KGAT [27]* : By introducing the concept of collaborative knowledge graphs, KGAT models high-order connectivity within the knowledge graph using attention mechanisms, improving the ability to capture deep associations between users and items.
- *KGIN [44]* : This method captures rich information from composite knowledge graphs by modeling relational user intents and performing path-aware aggregation.
- *SGL [32]* : Through random data augmentation operations combined with contrastive learning signals, SGL enhances graph-based collaborative filtering, improving the discrimination ability and robustness of graph-based recommendation models.
- *KGCL [13]* : By integrating knowledge graph information through self-supervised contrastive learning, KGCL effectively suppresses noise and tail effects, thereby improving the accuracy of recommendation systems.
- *MCCLK [14]* : This method performs hierarchical contrastive learning across global structures, local collaboration, and semantic views, self-supervised from user-item-entity graphs and their sub-graphs, to mine useful structural information.
- *KGRec [45]* : Combining generative and contrastive self-supervised learning, KGRec extracts key structures from collaborative filtering signals in the knowledge graph and uses a knowledge rationalization mechanism to suppress noise, improving recommendation performance.
- *DiffKG [46]* : By combining generative diffusion models and knowledge graph learning, DiffKG enhances the semantic and collaborative relationship modeling in collaborative filtering, significantly improving recommendation accuracy.
- *SMIN [47]* : Using semi-supervised learning to integrate multi-modal interactions, SMIN extracts significant sentiment features through both intra-modal and cross-modal modules, and integrates multi-modal information using hierarchical recurrent networks.
- *IHGCL [48]* : Leveraging meta-path-based contrastive learning to capture user-item intents in heterogeneous graphs and employing a bottlenecked autoencoder to suppress noise, enhancing recommendation accuracy.

Parameter Settings: To ensure a fair comparison, we applied uniform hyper-parameter settings for both the Diff-MSR framework and the baseline methods. Specifically, our DiffMSR model is implemented using PyTorch, with the Adam optimizer and the Xavier initializer. The training batch size is set to 1024, the embedding vector dimension is 64, the learning rate is set to a default value, and the number of GCN layers is 1. The L2 regularization decay is searched within the set $\{1e^{-1}, 1e^{-2}, 1e^{-3}, 1e^{-4}, 1e^{-5}, 1e^{-6}\}$. The loss weights λ_0 and λ_1 for different components are searched within $\{1, 0.5, 0.1, 0.01, 0.001\}$ and $\{1e^{-1}, 1e^{-2}, 1e^{-3}, 1e^{-4}, 1e^{-5}\}$, respectively. The hyper-parameter ω is searched within the range, and the temperature coefficient is searched within $\{0.02, 0.03, 0.05, 0.07, 0.1\}$. For the baseline methods, we use the same optimizer, parameter initializer, and batch size as DiffMSR. Shared hyper-parameters, such as the number of GCN layers and the L2 regularization term, are adjusted within the same range. For self-supervised methods, the temperature coefficient τ for contrastive learning is searched within $\{0.02, 0.03, 0.05, 0.07, 0.1\}$. For baseline methods using random information dropout, the dropout rate is adjusted within $\{0.1, \ldots, 0.9\}$.

B. Performance Comparison (RQ1)

We evaluated the overall performance of all methods, and the results are summarized in Table III and Fig. 4. To validate the significant performance improvement achieved by our DiffMSR model, the p-value is provided. From the evaluation results, we summarize the following key observations:

- Overall Performance Advantage of DiffMSR. Comprehensive experimental comparisons clearly indicate that the proposed model consistently outperforms all baseline approaches. DiffMSR effectively integrates multi-semantic information through cross-semantic contrastive learning and semantically-aware diffusion enhancement, leveraging a multi-semantic graph aggregation component. Specifically, our designed graph diffusion framework as a robust mechanism for graph generation, generating multi-semantics-aware interaction graphs through an iterative denoising process. This effectively increases the diversity and richness of the data, significantly improving recommendation accuracy. We also observe that better LLM embeddings provide more informative semantics, leading to more stable and accurate recommendation results, while

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TABLE III
PERFORMANCE COMPARISON OF ALL METHODS FOR RECALL@20, NDCG@20, PRECISION@20 AND HIT@20 ON THE DATASETS

Model	Recall@20	NDCG@20	Precision@20	Hit@20
LightGCN	0.3015	0.2007	0.0194	0.3383
KGCN	0.4407	0.2965	0.0264	0.4753
KGAT	0.5629	0.2812	0.0301	0.5769
CFKG	0.5456	0.3617	0.0286	0.5526
KGIN	0.5899	0.3538	0.0361	0.6498
SGL	0.4491	0.3596	0.0271	0.4895
KGCL	0.4614	0.3635	0.0277	0.4980
MCCLK	0.4911	0.3039	0.0297	0.5321
DiffKG	0.4449	0.2869	0.0251	0.4619
KGRec	0.6017	0.3555	0.0363	0.6633
SMIN	0.6222	0.4236	0.0360	0.6704
IHGCL	0.6293	0.3688	0.0315	0.6713
DiffMSR_ada	0.6427	0.4030	0.0361	0.6706
DiffMSR_3 large	**0.6694**	**0.4256**	**0.0378**	**0.6950**
Improve	6.37%	0.47%	4.13%	3.53%
p-value	$2.8e^{-6}$	$1.5e^{-7}$	$3.3e^{-8}$	$1.9e^{-7}$

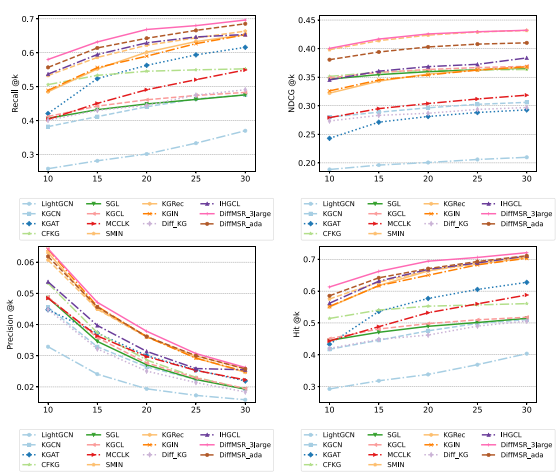

Fig. 4. Performance comparison of different methods in top-K recommendation.

less expressive embeddings may introduce minor semantic noise under sparse interactions. Experimental results show that DiffMSR achieves the best performance on all evaluation metrics, surpassing other models, regardless of whether OpenAI's GPT-3 large or ada-002 is used as the text encoder. These findings strongly validate the effectiveness and superiority of our model in integrating multi-source heterogeneous semantic information for enhanced recommendation quality.

- Improvement in Knowledge-Aware Recommendation Models. Previous methods, such as SGL, KGCL, and MCCLK, enhanced user-service interaction modeling through contrastive learning, but only achieved limited performance improvements compared to LightGCN. On the other hand, models that integrate knowledge graph information, such as KGAT, CFKG, and KGIN, have gained significant advantages in capturing user-service interaction patterns by leveraging entity relationships and attribute information from the knowledge graph. These models effectively alleviate the data sparsity problem faced by traditional collaborative filtering methods. Building upon these models, DiffMSR further enhances its denoising capability through a semantically-aware denoising network. It effectively identifies and filters out irrelevant relationships in the knowledge graph, resulting in more substantial improvements in recommendation quality. By intelligently incorporating semantic information and advanced denoising techniques, DiffMSR outperforms existing approaches, showcasing its ability to deliver highly accurate and personalized recommendations even in sparse data scenarios.
- Advantages of Combining Multi-Semantic and Generative Enhancement. KGRec excels at integrating knowledge graph semantic relevance and generative augmented self-supervised signals, effectively addressing the interaction sparsity issue. SMIN further surpasses KGRec in multi-modal information integration, highlighting the necessity of multi-semantic fusion. IHGCL further improves performance by capturing user-item intents through meta-path contrastive learning. DiffMSR achieves dynamic alignment and efficient integration of textual semantic features and graph structural features through the collaborative effect of semantically-aware contrastive view learning and multi-semantic generative graph diffusion models. Specifically, the semantically-aware contrastive view learning is responsible for aligning the representation spaces of different semantic views, while the multi-semantic graph diffusion model integrates multi-source semantic information into the generation of interaction graphs through an iterative denoising process. As illustrated in Fig. 5, the embedding distributions under structural, textual, and fused semantics reveal that DiffMSR produces more compact and globally consistent representations. This highlights its ability to effectively unify heterogeneous semantic signals into a single expressive space. DiffMSR outperforms existing methods in terms of recommendation accuracy, demonstrating the effectiveness of combining multi-semantic fusion with generative enhancement.

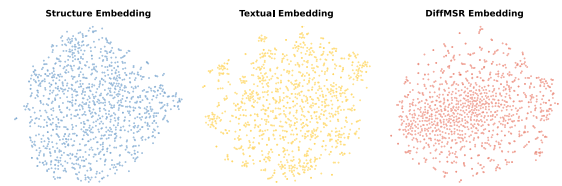

Fig. 5. Visualisation of the embedding distribution under different semantics.

C. Ablation Study (RQ2)

To evaluate the effectiveness of each component in the DiffMSR model, we conducted an ablation study using three different ablation variants:

- w/o ATT : In this variant, the attention module is removed to assess its impact on performance.
- w/o CL : In this variant, contrastive learning is removed to evaluate its contribution.
- w/o Diff : In this variant, the diffusion module is replaced with another generative model, VGAE, to assess the effectiveness of the designed diffusion module.

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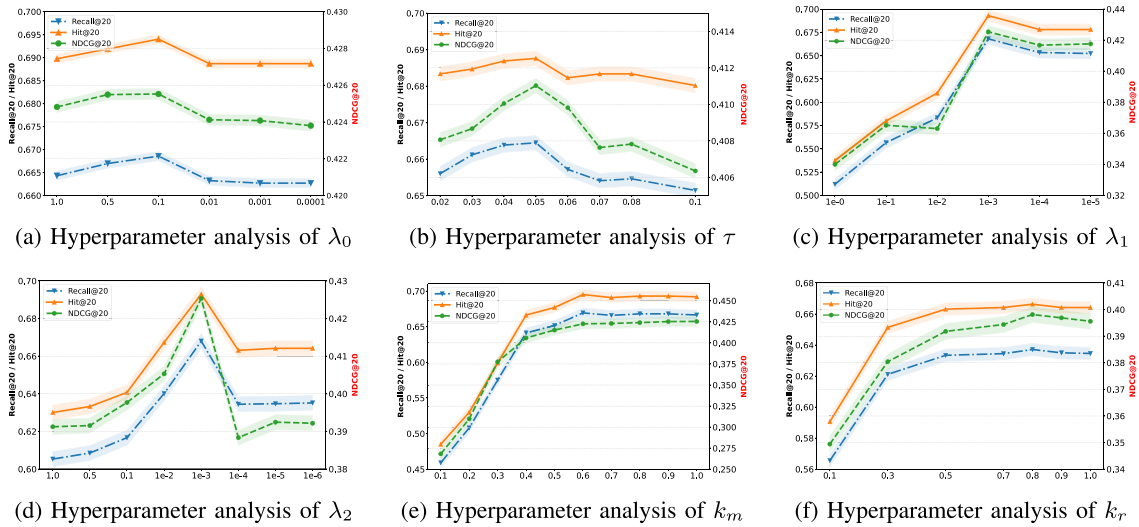

(a) Hyperparameter analysis of λ_0 (b) Hyperparameter analysis of τ (c) Hyperparameter analysis of λ_1

(d) Hyperparameter analysis of λ_2 (e) Hyperparameter analysis of k_m (f) Hyperparameter analysis of k_r

Fig. 6. Hyper-parameter study of the DiffMSR.

TABLE IV
ABLATION STUDIES ON KEY COMPONENTS OF DIFFMSR

Variants	Recall@20	NDCG@20	Precision@20	Hit@20
w/o ATT	0.6147	0.3819	0.0342	0.6450
w/o CL	0.6062	0.3488	0.0338	0.6365
w/o Diff	0.5967	0.3696	0.0328	0.6226
DiffMSR	**0.6694**	**0.4256**	**0.0378**	**0.6950**

TABLE V
IMPACT OF THE DIFFUSION STEP SIZE T ON PERFORMANCE AND TRAINING TIME

T	**Recall@20**	**NDCG@20**	**Precision@20**	**Hit@20**	**Time (s)**
$T = 2$	0.6496	0.4246	0.0367	0.6748	163.16
$T = 3$	0.6542	0.4228	0.0370	0.6791	181.30
$T = 4$	0.6558	0.4214	0.0371	0.6802	197.10
$T = 5$	**0.6694**	**0.4256**	**0.0378**	**0.6950**	222.51
$T = 6$	0.6627	0.4208	0.0374	0.6887	238.84
$T = 7$	0.6576	0.4203	0.0371	0.6855	250.02

The experimental results are reported in Table IV. From the results, we observe that removing the attention mechanism leads to a decline in performance across datasets, emphasizing its importance in distinguishing different nodes while capturing structural semantics. A similar trend is observed when contrastive learning is removed, highlighting its critical role in aligning textual and structural semantics. Finally, when the diffusion module is removed, the performance is at its lowest, validating the superiority of the diffusion model over alternative generative models, such as VGAE.

D. Hyperparameter Analysis (RQ3)

We investigated the sensitivity of key hyperparameters in our DiffMSR. The results of this analysis are presented in Table V, Figs. 6 and 7, which provide a comprehensive overview of the findings.

Impact of the diffusion step size T in the Graph Diffusion Model. The diffusion step size T determines how many times the interaction data are perturbed and denoised. As shown in Table V, model performance improves with increasing T,

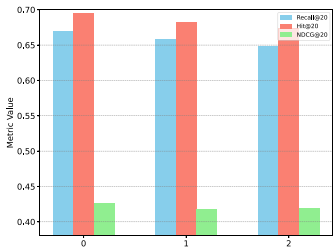

Fig. 7. Comparison of the effect of different semantic alignments.

reaching a peak at $T = 5$. This suggests that a moderate number of diffusion steps helps capture richer semantics and suppress noise. However, further increasing T yields no significant gains while incurring additional computational cost. Therefore, we set T to 5 achieve a good balance between effectiveness and efficiency.

Impact of Key Hyperparameter λ_0 in the Graph Diffusion Model. The hyperparameter λ_0 plays a pivotal role in determining the intensity of specific semantic signal injection (MSI) and guiding the generation of semantically aware user-service interaction graphs in the multi-semantic graph diffusion model. We experimented with values from the range $\{1.0, 0.5, 0.1, 0.01, 0.001, 0.0001\}$, and as shown in Fig. 6(a), the model achieves optimal performance when λ_0 is set to 0.1. These findings highlight the importance of selecting an appropriate value of λ_0 to facilitate DiffMSR in constructing semantically aware user-service interaction graphs, thereby enhancing recommendation performance.

Impact of τ and λ_1 in Cross-Semantic Data Augmentation. In cross-semantic contrastive learning, the hyperparameters τ (temperature coefficient) and λ_1 (weight of InfoNCE loss) are critical. As shown in Fig. 6(b) and (c), the best performance is clearly seen to be obtained when $\tau = 0.05$ and $\lambda_1 = 1e - 3$. This observation underscores the substantial impact of cross-semantic contrastive learning on the effectiveness of our DiffMSR framework.

Impact of Weight k_m and Regularization Term λ_2. Since different semantics contribute differently to the final multi-semantic representation, k_m is used to control the weight of each semantic during the aggregation process. As shown in

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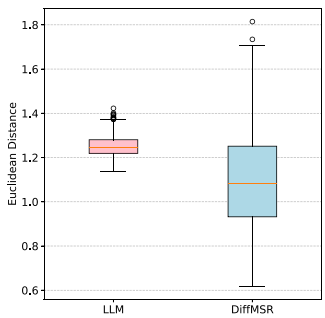


Fig. 8. Visual representation before and after alignment with KGAT as the backbone model.

Fig. 6(e), the model achieves optimal performance when k_m is set to 0.6, further confirming the importance of aggregating various semantic information. Furthermore, the regularization term λ_2 plays a critical role in preventing model overfitting. As illustrated in Fig. 6(d), the model performs best when λ_2 is set to $1e-3$.

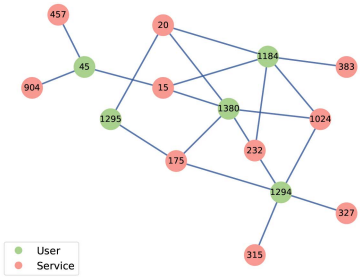
Impact of the retention rate of the edge on the diffusion process. During the diffusion process, a certain proportion of edges needs to be retained to preserve the original graph structure as much as possible, ensuring that critical information is not lost when Gaussian noise is introduced. As shown in Fig. 6(f), the model achieves optimal performance when the edge retention ratio (k_r) is set to 0.8.

Impact of Multi-Semantic Alignment Methods. To align different raw semantic feature embeddings, we introduced three alignment methods: '0: R∗R'which uses a parameter matrix; '1: Linear' which is based on linear transformation; and '2: Allrecipes' a hybrid approach that aligns knowledge graph semantic features using a parameter matrix while employing linear mapping for service text semantic features. As shown in Fig. 7, the alignment method that utilizes the parameter matrix demonstrated the best performance in the service data set.

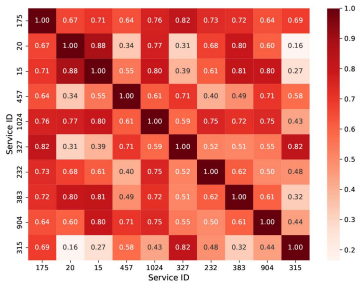
Visualization of bridging the gap. To investigate the impact of aligning different semantic features on bridging the gap between diverse representation spaces, we captured service representations from the backbone network (KG), LLM, and our proposed model. We calculated the distances between the embedding vectors of the services obtained from the backbone network and those of the LLM and DiffMSR. To ensure experimental reliability, we selected KGAT as the backbone network. As shown in Fig. 8, the results are visualized as box plots. DiffMSR effectively reduced the gap between different representation spaces, significantly enhancing the representation capability of the backbone model, while the right plot illustrates the distance between DiffMSR and the backbone model. Initially, the vectors from GNN and LLM exhibited large euclidean distances on the left. However, after applying semantic feature alignment, the distances between the vectors were significantly reduced. These findings confirm the effectiveness of DiffMSR in aligning structural and textual representations across diverse spaces.

E. Model Interpretability Study (RQ4)

To evaluate the generative capability of our proposed DiffMSR framework, we introduced a Semantic Signal Injection (MSI) mechanism and performed a detailed case analysis on the generated user-service interaction graph. Fig. 9(a) illustrates a randomly sampled subgraph of user-service interactions from the service dataset. Fig. 9(b) shows the service similarity based on multi-semantic feature representation.



(a) Semantic-Aware User-Service Graph.



(b) Semantic Feature Similarity Heatmap.

Fig. 9. Case study on the generated semantic-aware user-service graph.

The results reveal a strong correlation between the constructed graph structure and the similarities based on semantic features. For example, in the generated graph, Service 20 and Service 15 are neighbors of User 1380, showing a high similarity score of 0.88 on the heat map, which represents the highest score between these two services. Similarly, Services 175 and 327, with a similarity score of 0.82, are both connected to User 1294. Services with similarity scores greater than 0.8 are often connected to the same user, indicating semantic-aware similarities between them. In particular, these services are not directly connected in the original user-service interaction graph, and their relationships were established through semantic feature integration.

VI. Discussion

While DiffMSR builds upon established components including large language models, knowledge graphs, and contrastive learning, its fundamental contribution lies in the principled unification of these elements within a diffusion-based generative framework specifically designed for service recommendation. This integration introduces a novel paradigm where multi-semantic representations undergo progressive refinement through controlled denoising processes, enabling the model to iteratively enhance user and service embeddings while mitigating noise inherent in heterogeneous semantic sources.

In particular, the semantic-aware graph diffusion process bridges textual semantics and structured knowledge, which are traditionally modeled independently. Unlike conventional hybrid models that process multimodal signals in isolation, DiffMSR performs iterative cross-semantic contrastive learning and residual enhanced aggregation, effectively aligning semantic spaces and strengthening higher-order collaborative signals. This unified design moves beyond pipeline-style fusion and provides a novel solution to semantic sparsity and imbalance.

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VII. Conclusion and Future Work

Following the above discussions and insights, in this paper, we propose DiffMSR, a multi-semantic graph diffusion model for service recommendation, which innovatively employs a diffusion generation framework to reconstruct user-service interaction graphs, effectively addressing data sparsity and noise issues. By integrating structural features from knowledge graphs with textual semantic representations from large language models, the model significantly enhances service representation capabilities. In addition, it implements cross-semantic contrastive learning with dynamic weights to achieve effective alignment of heterogeneous information. The experimental results demonstrate that the model outperforms the best baseline of 4.13% and 6.37% in terms of recommendation accuracy and recall, respectively. The superior performance of DiffMSR demonstrates its effectiveness in alleviating the sparsity issue and reducing the impact of interaction noise, thereby enabling more accurate and reliable service recommendations.

This paper opens up a new research direction for service recommendation systems by integrating diffusion models with multi-semantic representations. Following this direction, several promising ideas are worth further exploration: First, knowledge graphs typically contain numerous irrelevant entities and facts, which not only increases computational overhead, but also affects model performance. Research could focus on how to utilize diffusion mechanisms to learn optimal propagation paths, achieve efficient information filtering, and incorporate attention mechanisms to enhance the identification of key information. Second, incorporating Quality of Service (QoS) constraints as conditional information into the diffusion process shows great potential. This approach not only ensures semantic relevance in reasoning paths, but also meets specific service quality requirements, thereby enhancing the practical value of service recommendation systems. Meanwhile, we plan to explore lightweight variants of the model via techniques such as model distillation and pruning to further improve scalability in constrained settings.

References

[1] Z. Zheng, X. Li, M. Tang, F. Xie, and M. R. Lyu, "Web service QoS prediction via collaborative filtering: A survey," *IEEE Trans. Services Comput.*, vol. 15, no. 4, pp. 2455–2472, Jul./Aug. 2022.

[2] K. Mecheri, S. Klai, and L. Souici-Meslati, "Deep learning based Web service recommendation methods: A survey," *J. Intell. Fuzzy Syst.*, vol. 44, no. 6, pp. 9879–9899, 2023.

[3] Y. Zhao and J. Chen, "Vector-indistinguishability: Location dependency based privacy protection for successive location data," *IEEE Trans. Comput.*, vol. 73, no. 4, pp. 970–979, Apr. 2024.

[4] B. Cao et al., "Web service recommendation via integrating heterogeneous graph attention network representation and FiBiNET score prediction," *IEEE Trans. Services Comput.*, vol. 16, no. 5, pp. 3837–3850, Sep./Oct. 2023.

[5] B. Cao, Q. Peng, X. Xie, Z. Peng, J. Liu, and Z. Zheng, "Web service recommendation via combining topic-aware heterogeneous graph representation and interactive semantic enhancement," *IEEE Trans. Services Comput.*, vol. 17, no. 6, pp. 4451–4466, Nov./Dec. 2024.

[6] M. Shi, Y. Zhuang, Y. Tang, M. Lin, X. Zhu, and J. Liu, "Web service network embedding based on link prediction and convolutional learning," *IEEE Trans. Services Comput.*, vol. 15, no. 6, pp. 3620–3633, Nov./Dec. 2022.

[7] P. Bonte, S. V. Hautte, F. De Turck, S. Van Hoecke, and F. Ongenae, "Benchmarking whole knowledge graph embedding techniques," *Int. J. Softw. Eng. Knowl. Eng.*, vol. 34, no. 1, pp. 163–184, 2024.

[8] B. Cao et al., "PRKG: Pre-training representation and knowledge-graph-enhanced Web service recommendation for mashup creation," *IEEE Trans. Netw. Service Manag.*, vol. 21, no. 2, pp. 1737–1749, Apr. 2024.

[9] H. Wu, Y. Duan, K. Yue, and L. Zhang, "Mashup-oriented web API recommendation via multi-model fusion and multi-task learning," *IEEE Trans. Services Comput.*, vol. 15, no. 6, pp. 3330–3343, Nov./Dec. 2022.

[10] J. Chen, H. Dong, X. Wang, F. Feng, M. Wang, and X. He, "Bias and Debias in recommender system: A survey and future directions," *ACM Trans. Inf. Syst.*, vol. 41, no. 3, pp. 1–39, 2023.

[11] X. He, K. Deng, X. Wang, Y. Li, Y. Zhang, and M. Wang, "LightGCN: Simplifying and powering graph convolution network for recommendation," in *Proc. 43rd Int. ACM SIGIR Conf.*, 2020, pp. 639–648.

[12] J. Wu et al., "Self-supervised graph learning for recommendation," in *Proc. 44th Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2021, pp. 726–735.

[13] Y. Yang, C. Huang, L. Xia, and C. Li, "Knowledge graph contrastive learning for recommendation," in *Proc. 45th Int. ACM SIGIR Conf.*, 2022, pp. 1434–1443.

[14] D. Zou et al., "Multi-level cross-view contrastive learning for knowledge-aware recommender system," in *Proc. 45th Int. ACM SIGIR Conf.*, 2022, pp. 1358–1368.

[15] L. Xia, C. Huang, Y. Xu, J. Zhao, D. Yin, and J. Huang, "Hypergraph contrastive collaborative filtering," in *Proc. 45th Int. ACM SIGIR Conf.*, 2022, pp. 70–79.

[16] A. Abibullayeva, H. Kılıç, and A. Cetin, "An ensemble keyword extraction model for news texts with statistical and graphical features," *Int. J. Softw. Eng. Knowl. Eng.*, vol. 34, no. 7, pp. 1047–1061, 2024.

[17] M. Shi, J. Liu, D. Zhou, M. Tang, and B. Cao, "WE-LDA: A word embeddings augmented LDA model for web services clustering," in *Proc. IEEE Int. Conf. Web Serv.*, 2017, pp. 9–16.

[18] J. Achiam et al., "GPT-4 technical report," 2023, *arXiv:2303.08774*.

[19] J. Harte, W. Zorgdrager, P. Louridas, A. Katsifodimos, D. Jannach, and M. Fragkoulis, "Leveraging large language models for sequential recommendation," in *Proc. 17th ACM Conf. Recommender Syst.*, 2023, pp. 1096–1102.

[20] L. Wu et al., "A survey on large language models for recommendation," *World Wide Web*, vol. 27, no. 5, 2024, Art. no. 60.

[21] W. Wei et al., "LLMRec: Large language models with graph augmentation for recommendation," in *Proc. 17th ACM Int. Conf. Web Search Data Mining*, 2024, pp. 806–815.

[22] X. Ren et al., "Representation learning with large language models for recommendation," in *Proc. ACM Web Conf.*, 2024, pp. 3464–3475.

[23] M. Zhang et al., "GraphTranslator: Aligning graph model to large language model for open-ended tasks," in *Proc. ACM Web Conf.*, 2024, pp. 1003–1014.

[24] S. H. Ghafouri, S. M. Hashemi, and P. C. K. Hung, "A survey on Web service QoS prediction methods," *IEEE Trans. Services Comput.*, vol. 15, no. 4, pp. 2439–2454, Jul./Aug. 2022.

[25] R. Xiong, J. Wang, N. Zhang, and Y. Ma, "Deep hybrid collaborative filtering for web service recommendation," *Expert Syst. Appl.*, vol. 110, pp. 191–205, 2018.

[26] B. Cao et al., "Web API recommendation via combining graph attention representation and deep factorization machines quality prediction," *Concurrency Comput. Pract. Exp.*, vol. 34, no. 21, 2022, Art. no. e7069.

[27] X. Wang, X. He, Y. Cao, M. Liu, and T.-S. Chua, "KGAT: Knowledge graph attention network for recommendation," in *Proc. 25th ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2019, pp. 950–958.

[28] X. Wang, X. Liu, J. Liu, X. Chen, and H. Wu, "A novel knowledge graph embedding based API recommendation method for mashup development," *World Wide Web*, vol. 24, no. 3, pp. 869–894, 2021.

[29] H. Mezni, D. Benslimane, and L. Bellatreche, "Context-aware service recommendation based on knowledge graph embedding," *IEEE Trans. Knowl. Data Eng.*, vol. 34, no. 11, pp. 5225–5238, Nov. 2022.

[30] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.

[31] Y. You, T. Chen, Y. Sui, T. Chen, Z. Wang, and Y. Shen, "Graph contrastive learning with augmentations," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2020, pp. 5812–5823.

[32] L. Procopio, R. Tripodi, and R. Navigli, "SGL: Speaking the graph languages of semantic parsing via multilingual translation," in *Proc. 2021 Conf. North Amer. Chapter Assoc. Comput. Linguistics: Hum. Lang. Technol.*, 2021, pp. 325–337.

[33] J. Yu, H. Yin, X. Xia, T. Chen, L. Cui, and Q. V. H. Nguyen, "Are graph augmentations necessary? Simple graph contrastive learning for recommendation," in *Proc. 45th ACM SIGIR Conf.*, 2022, pp. 1294–1303.

[34] X. Cai, C. Huang, L. Xia, and X. Ren, "LightGCL: Simple yet effective graph contrastive learning for recommendation," 2023, *arXiv:2302.08191*.

[35] J. Choi and D. Ryu, "QoS-aware graph contrastive learning for Web service recommendation," in *Proc. 30th Asia-Pacific Softw. Eng. Conf.*, 2023, pp. 171–180.

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[36] D. Yu, T. Yu, D. Wang, and S. Wang, "Long tail service recommendation based on cross-view and contrastive learning," *Expert Syst. Appl.*, vol. 238, 2024, Art. no. 121957.
[37] T. Brown et al., "Language models are few-shot learners," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2020, pp. 1877–1901.
[38] J. Gong et al., "A population-to-individual tuning framework for adapting pretrained LM to on-device user intent prediction," in *Proc. 30th ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2024, pp. 896–907.
[39] Z. Chen et al., "Label-free node classification on graphs with large language models (LLMs)," 2023, *arXiv:2310.04668*.
[40] W. Wang, Y. Xu, F. Feng, X. Lin, X. He, and T.-S. Chua, "Diffusion recommender model," in *Proc. 46th ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2023, pp. 832–841.
[41] Y. Jiang, L. Xia, W. Wei, D. Luo, K. Lin, and C. Huang, "DiffMM: Multimodal diffusion model for recommendation," in *Proc. 32nd ACM Int. Conf. Multimedia*, 2024, pp. 7591–7599.
[42] Y. Zhang, Q. Ai, X. Chen, and P. Wang, "Learning over knowledge-base embeddings for recommendation," 2018, *arXiv: 1803.06540*.
[43] R. Kojima, S. Ishida, M. Ohta, H. Iwata, T. Honma, and Y. Okuno, "KGCN: A graph-based deep learning framework for chemical structures," *J. Cheminformatics*, vol. 12, pp. 1–10, 2020.
[44] X. Wang et al., "Learning intents behind interactions with knowledge graph for recommendation," in *Proc. Web Conf.*, 2021, pp. 878–887.
[45] Y. Yang, C. Huang, L. Xia, and C. Huang, "Knowledge graph self-supervised rationalization for recommendation," in *Proc. 29th ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2023, pp. 3046–3056.
[46] Y. Jiang, Y. Yang, L. Xia, and C. Huang, "DiffKG: Knowledge graph diffusion model for recommendation," in *Proc. 17th ACM Int. Conf. Web Search Data Mining*, 2024, pp. 313–321.
[47] Z. Lian, B. Liu, and J. Tao, "SMin: Semi-supervised multi-modal interaction network for conversational emotion recognition," *IEEE Trans. Affect. Comput.*, vol. 14, no. 3, pp. 2415–2429, Third Quarter 2023.
[48] L. Sang, Y. Wang, Y. Zhang, Y. Zhang, and X. Wu, "Intent-guided heterogeneous graph contrastive learning for recommendation," *IEEE Trans. Knowl. Data Eng.*, vol. 37, no. 4, pp. 1915–1929, Apr. 2025.
[49] M. Li, Z. Li, C. Huang, Y. Jiang, and X. Wu, "EduGraph: Learning path-based hypergraph neural networks for MOOC course recommendation," *IEEE Trans. Big Data*, vol. 10, no. 6, pp. 706–719, Dec. 2024.
[50] M. Li, A. Micheli, Y. G. Wang, S. Pan, P. Lió, and G. S. Gnecco, "Guest editorial: Deep neural networks for graphs: Theory, models, algorithms, and applications," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 4, pp. 4367–4372, Apr. 2024.

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